

Advertising, Social Media and Parasocial Recommendations

New Influencer-Centered Modeling Approach

Julian Klix | **Preliminary Draft**

July, 2026



Model Setup: The Product Landscape

- Multiple product categories
 - ⇒ denote $k \in \{A, B, \dots\}$
 - ⇒ will allow for influencer “specialization”
 - ⇒ relevant only for informational purposes

Not yet used right now

- Multiple quality dimensions
 - ⇒ denote $j \in \{1, 2, \dots\}$
 - ⇒ relevant for realized utility

Model Setup: The Firm Side

The Monopolist

- Single firm with high (h) or low (l) quality in each dimension [generically q_j]
- Realization of all q_j are private information with $P[q_j = H] \equiv \pi$
- Fix $|J| = 2$ gives three classes q : H (both $q_j = h$), M (one $q_j = h$) or L (no $q_j = h$)
- Produced at constant, homogeneous marginal cost c
- Competes (indirectly) against outside option
- Can advertise product on social media at exogenous cost κ

The Outside Option

- Available instead of purchasing the product
⇒ Can be thought of as background Bertrand competition with cost $c_0 = v_0 + p_0$

Model Setup: The Consumer Side

- One (representative) consumer with unit demand
- Randomly drawn product dimension type $\theta \in J$
- Receives utility v_0 from outside option and v_q from the new product
 - ⇒ utility of product given by $u_c(\theta) = v_{q_\theta}$ [i.e. quality of dimension θ]
 - ⇒ normalize $v_l = 0$, simplify notation $v_h = v$
- Can obtain **one** recommendation r prior to purchase
 - ⇒ either from environment (non-strategic) or social media (potent. strategic)

Assumption 1

The outside option v_0 is sufficiently large to dominate random purchase. That is, $v_0 > \pi v - p$.

Model Setup: The Influencer Side

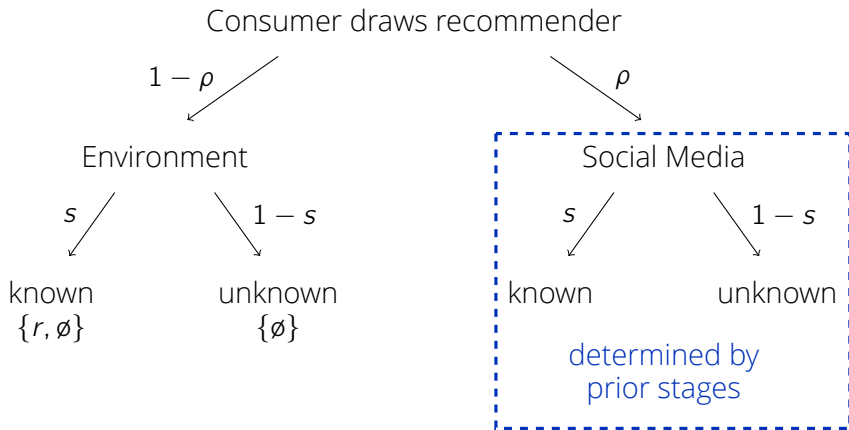
- One (representative) Influencer that can give recommendations
 - Product dimension type $\iota \in J$
 - direct utility from payment $\Pi_{\mathcal{I}}$, indirect utility from value-added to consumer
 $\Rightarrow$ altruism parameter α captures trade-off, distr. following (cont.) F_{α}

$$u_{\mathcal{I}} = (1 - \alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - u_c(\emptyset)] = (1 - \alpha)\Pi_{\mathcal{I}} + \alpha[\mathbb{E}[v_{\theta}|r] - p - v_0]$$
 $\Rightarrow$ non-strategic environment encapsulated by $\alpha = 1$
- Type-match probability ("similarity") $\sigma \Rightarrow$ potentially σ_0, σ_I

Parasocial Relationships

The consumer's parasocial relationship is captured by overestimating the influencer's similarity, i.e., $\sigma_{\eta} > \sigma$. [Specific form to be determined]

Model Setup: The Recommendation Procedure



Model Setup: Timing

1. Fundamentals are drawn ($p, \kappa, q, v_j, \dots$)
2. Firms learn q , knowing recommenders learn v_ℓ
3. Firm decides on endorsement offer
4. Influencer decides on accepting endorsement offer
5. Consumer draws recommender
6. Consumer observes potential recommendation
 $\Rightarrow$ can distinguish genuine from paid recommendations
7. Consumer makes purchase decision
 $\Rightarrow$ utilities are realized

An Endorsement-Free Benchmark

- Outside option of $v_0 \Rightarrow$ by assumption, obtained without recommendation
- Consumer updates belief about v_θ after receiving a recommendation: $\pi \xrightarrow{r} \pi_r$

$$\pi_r \equiv \mathbb{P}[v_\theta = v | r] = \sigma \frac{1 \cdot \mathbb{P}[v]}{1 \cdot \mathbb{P}[v] + 0 \cdot \mathbb{P}[0]} + (1 - \sigma) \frac{\pi \cdot \mathbb{P}[v]}{\pi \cdot \mathbb{P}[v] + \pi \cdot \mathbb{P}[0]} = \sigma + (1 - \sigma)\pi$$

$\Rightarrow$ assume expected utility $\pi_r v - p = \pi v + (1 - \pi)\sigma v - p > v_0$

- without endorsements ($\Pi_{\mathcal{I}} = 0$), influencer's interests perfectly aligned
 $\Rightarrow$ recommend a (known) product iff $v_i = v$

Equilibrium of the Endorsement-Free Benchmark

Without endorsements, there exists an equilibrium in which any high-value product gets recommended and all recommendations are followed.

The HML-Endorsement Equilibrium

HML-Endorsement Equilibrium

For sufficiently small κ there exists an equilibrium with endorsements in which

- (a) All firm types ($q = H, M, L$) decide to seek endorsements at cost κ .
- (b) There exist thresholds $\bar{\alpha}_s$ and $\bar{\alpha}_{\bar{s}}$ such that an influencer
 - i. with $\alpha \leq \bar{\alpha}_{\bar{s}}$ endorses the product unseen.
 - ii. with $\alpha \leq \bar{\alpha}_s$ endorses any product and $\alpha > \bar{\alpha}_s$ only $v_l = v$ products.
- (c) All consumers follow genuine recommendations.
- (d) Sufficiently parasocial consumers ($\sigma_\eta > \bar{\sigma}$) also follow endorsements.

- Equilibrium with full firm-type pooling

Firm Decision

- Firm endorses if costs are sufficiently small $\kappa \leq (D_E^q - D_0^q)(p - c)$
 $\Rightarrow$ effective demand with (D_E^q) and without (D_0^q) endorsements depends on v_ℓ
- Denote share of parasocial customers ($\sigma_\eta > \bar{\sigma}$) by ψ_η

v_ℓ	D_0	D_E
v	s	$\rho(1 - s)\psi_\eta F_\alpha(\bar{\alpha}_s) - \rho s(1 - \psi_\eta) + s$
0	0	$\rho(1 - s)\psi_\eta F_\alpha(\bar{\alpha}_s) + \rho s\psi_\eta F_\alpha(\bar{\alpha}_s)$

- Note that absolute gains $(D_E^q - D_0^q)$ decrease with q_ℓ (and hence with q)
 $\Rightarrow$ all firms profit iff $\kappa \leq \bar{\kappa}_H \equiv \rho[\psi_\eta(1 - s)F_\alpha(\bar{\alpha}_s) - (1 - \psi_\eta)s](p - c)$
 $\Rightarrow$ Note: $\bar{\kappa}_H > 0 \iff \frac{\psi_\eta}{1 - \psi_\eta} F_\alpha(\bar{\alpha}_s) > \frac{s}{1 - s} \rightarrow$ necessary cond. for HML-eq.

Consumer Decision

- Reminder: Consumers can differentiate paid from genuine recommendations
 $\Rightarrow$ paid ones can still be **aligned** (i.e., iff $v_\ell = v$) for high $\alpha \Rightarrow$ denote prob. ψ_α
- Genuine recommendations always aligned (Benchmark) $\Rightarrow$ always followed
- Follow paid recommendation iff v_0 or p sufficiently small and

$$\mathbb{E}[v_\theta | r^p] = \psi_\alpha(\pi + (1 - \pi)\sigma)v + (1 - \psi_\alpha)\pi v = [\pi + (1 - \pi)\sigma\psi_\alpha]v \geq v_0 + p$$

$\Rightarrow$ satisfied for sufficiently parasocial consumers $\Rightarrow \sigma_\eta \geq \bar{\sigma} \equiv \frac{v_0 + p}{(1 - \pi)\psi_\alpha v} - \frac{\pi}{(1 - \pi)\psi_\alpha}$

$\Rightarrow$ denote probability of following paid recommendations by ψ_η

Influencer Decision

- Recall influencer's utility: $u_{\mathcal{I}} = (1 - \alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - v_0]$
- Three influencer types, accept endorsement iff
 - $\bar{s}$: $(1 - \alpha)\kappa + \alpha[\pi v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \pi v + p + v_0}$
 - $s|v$: $(1 - \alpha)\kappa + \alpha[(1 - \pi)\sigma v + \pi v - p - v_0] > (1 - \alpha)\kappa \geq 0$
 $\Rightarrow$ always true (interests perfectly aligned)
 - $s|0$: $(1 - \alpha)\kappa + \alpha[(1 - \sigma)\pi v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_s \equiv \frac{\kappa}{\kappa - (1 - \sigma)\pi v + p + v_0}$

$\Rightarrow$ aligned paid recommendation probability $\psi_{\alpha} = \frac{\pi s(1 - F_{\alpha}(\bar{\alpha}_s))}{s(\pi + (1 - \pi)F_{\alpha}(\bar{\alpha}_s)) + (1 - s)F_{\alpha}(\bar{\alpha}_{\bar{s}})}$

Comparative Statics

Influencer

- $\bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \pi v + p + v_0}$
 $\bar{\alpha}_s \equiv \frac{\kappa}{\kappa - (1 - \sigma)\pi v + p + v_0}$
 $\Rightarrow$ incr. in κ, π, v decr. in p, v_0 incr. in κ, π, v decr. in p, v_0, σ
- $\psi_{\alpha} \equiv \frac{\pi s(1 - F_{\alpha}(\bar{\alpha}_s))}{s(\pi + (1 - \pi)F_{\alpha}(\bar{\alpha}_s)) + (1 - s)F_{\alpha}(\bar{\alpha}_{\bar{s}})}$
 $\Rightarrow$ incr. in s, p, v_0, σ decr. in κ, v and $\bar{\alpha}$ ambiguous in π

Consumer

- $\bar{\sigma} \equiv \frac{v_0 + p - \pi v}{(1 - \pi)v\psi_{\alpha}}$
 $\Rightarrow$ incr. in κ decr. in ψ_{α}, s, σ

Firm

- $\bar{\kappa}_H \equiv \rho[\psi_{\eta}(1 - s)F_{\alpha}(\bar{\alpha}_{\bar{s}}) - (1 - \psi_{\eta})s](p - c)$
 $\Rightarrow$ incr. in ρ decr. in c

Consumer Surplus & Welfare

- How to define welfare?

⇒ p, κ welfare neutral → transform influencer utility

$$W = E[v|r] - p + p - c - \kappa + (1 - \alpha)\kappa + \alpha(E[v|r] - p - v_0)$$

$$\rightarrow W = E[v|r] - c + \frac{\alpha}{1-\alpha}(E[v|r] - p - v_0)$$

- Focus on (net) consumer surplus $CS - v_0$ ⇒ multiple components

$$r_g: \quad [\sigma + (1 - \sigma)\pi]v - p - v_0 \quad \text{with prob. } (1 - \rho)s\pi$$

$$r_p|\bar{s}: \quad \pi v - p - v_0 \quad \text{with prob. } \rho(1 - s)\psi_\eta F_\alpha(\bar{\alpha}_s)$$

$$r_p|0: \quad (1 - \sigma)\pi v - p - v_0 \quad \text{with prob. } \rho s(1 - \pi)\psi_\eta F_\alpha(\bar{\alpha}_s)$$

$$r_p|v: \quad [\sigma + (1 - \sigma)\pi]v - p - v_0 \quad \text{with prob. } \rho s\pi\psi_\eta$$

⇒ Trivially, CS lower with endorsements

⇒ Effect of ψ_η ambiguous (increases “misleading” & aligned endorsement)

Example: HML-Equilibrium

Parameters

- $v = 6, v_0 = 2, p = 3, \kappa = 0.5, \pi = 0.8$
- $\rho = 0.7, s = 0.2, \sigma = 0.8$
- $\alpha \sim U[0, 1] \quad \sigma_\eta = \eta + (1 - \eta)\sigma, \eta \sim U[0, 1]$

Equilibrium Quantities

- $\bar{\alpha}_{\bar{s}} \approx 0.71 \quad \bar{\alpha}_s \approx 0.11$
 $\Rightarrow \psi_\alpha \approx 0.19$
- $\bar{\sigma} \approx 0.86$
 $\Rightarrow \psi_\eta \approx 0.69$
- $\bar{\kappa}_H \approx 0.7$

How to best visualize Examples?

An Interesting Observation: The Non-Parasocial Equilibrium

Non-Parasocial Equilibria

There exist HML-Equilibria for which $\bar{\sigma} \leq \sigma$. That is, all consumers follow paid recommendation, even in the absence of a parasocial component (i.e., for a degenerate σ_η -distribution).

- Usually accompanied by low $\bar{\alpha} \Rightarrow$ high alignment probability ψ_α
- Collective reputation of endorsing influencers good enough
- Consumer surplus still lower than without endorsements

$\Rightarrow$ Non-Parasocial Equilibria not limited to HML-Equilibria

The (first) ML-Endorsement Equilibrium

- Two (mutually exclusive) equilibria classes dependent on $\sigma v - p \gtrless v_0$
 $\Rightarrow$ focus (first) on $\sigma v - p > v_0$ [genuine recommendation for M -types]

ML-Endorsement Equilibrium (Aligned Influencers)

For intermediate κ there exists an equilibrium with endorsements in which

- Only firm types $q = M, L$ decide to use endorsements at cost κ .
- There exist thresholds $\bar{\alpha}_s$ and $\bar{\alpha}_{\bar{s}}$ such that an influencer
 - with $\alpha \leq \bar{\alpha}_{\bar{s}}$ endorses the product unseen.
 - with $\alpha \leq \bar{\alpha}_s$ endorses any product and $\alpha > \bar{\alpha}_s$ only $v_L = v$ products.
- All consumers follow genuine recommendations.
- Sufficiently parasocial consumers ($\sigma_\eta > \bar{\sigma}$) also follow endorsements.

Firm Decision

- Firm endorses if costs are sufficiently small $\kappa \leq (D_E^q - D_0^q)(p - c)$
 $\Rightarrow$ effective demand with (D_E^q) and without (D_0^q) endorsements depends on v_ℓ
- Denote share of parasocial customers ($\sigma_\eta > \bar{\sigma}$) by ψ_η

v_ℓ	D_0	D_E
v	s	$\rho(1 - s)\psi_\eta F_\alpha(\bar{\alpha}_s) - \rho s(1 - \psi_\eta) + s$
0	0	$\rho(1 - s)\psi_\eta F_\alpha(\bar{\alpha}_s) + \rho s\psi_\eta F_\alpha(\bar{\alpha}_s)$

- Note that absolute gains $(D_E^q - D_0^q)$ decrease with q_ℓ (and hence with q)
 $\Rightarrow$ H -type firm not endorsing iff $\kappa > \rho[\psi_\eta(1 - s)F_\alpha(\bar{\alpha}_s) - (1 - \psi_\eta)s](p - c)$
 $\Rightarrow$ others endorse iff

$$\kappa \leq \bar{\kappa}_M \equiv \rho[\psi_\eta(1 - s)F_\alpha(\bar{\alpha}_s) - \pi(1 - \psi_\eta)s + (1 - \pi)\psi_\eta s F_\alpha(\bar{\alpha}_s)](p - c)$$

Consumer Decision

- Reminder: Consumers can differentiate paid from genuine recommendations
 $\Rightarrow$ paid ones can still be **aligned** (i.e., iff $v_\ell = v$) for high $\alpha \Rightarrow$ denote prob. ψ_α
- Expected value from unaligned recommendations: $\frac{\pi(1-\pi)}{1-\pi^2} v = \frac{\pi}{1+\pi} v$
- Genuine recommendations always aligned (Benchmark) $\Rightarrow$ always followed
- Follow paid recommendation iff v_0 or p sufficiently small and

$$\mathbb{E}[v_\theta | r^p] = \psi_\alpha \sigma v + (1 - \psi_\alpha) \frac{\pi}{1+\pi} v \geq v_0 + p$$

$\Rightarrow$ satisfied for sufficiently parasocial consumers $\Rightarrow \sigma_\eta \geq \bar{\sigma} \equiv \frac{v_0+p}{\psi_\alpha v} - \frac{\pi(1-\psi_\alpha)}{(1+\pi)\psi_\alpha}$

$\Rightarrow$ denote probability of following paid recommendations by ψ_η

Influencer Decision

- Recall influencer's utility: $u_{\mathcal{I}} = (1 - \alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - v_0]$
 - Three influencer types, accept endorsement iff
 - $\bar{s}$: $(1 - \alpha)\kappa + \alpha[\frac{\pi}{1+\pi}v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \frac{\pi}{1+\pi}v + p + v_0}$
 - $s|v$: $(1 - \alpha)\kappa \geq 0$
 $\Rightarrow$ always true (interests perfectly aligned as long as $\sigma v - p \geq v_0$)
 - $s|0$: $(1 - \alpha)\kappa + \alpha[(1 - \sigma)\pi v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_s \equiv \frac{\kappa}{\kappa - (1 - \sigma)\pi v + p + v_0}$
- $\Rightarrow$ aligned paid recommendation prob. $\psi_{\alpha} = \frac{\pi s(1 - F_{\alpha}(\bar{\alpha}_s))}{s(\pi + F_{\alpha}(\bar{\alpha}_s)) + (1 + \pi)(1 - s)F_{\alpha}(\bar{\alpha}_{\bar{s}})}$

Comparative Statics

Influencer

- $\bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \frac{\pi}{1+\pi}v + p + v_0}$
 $\bar{\alpha}_s \equiv \frac{\kappa}{\kappa - (1-\sigma)\pi v + p + v_0}$
 $\Rightarrow$ incr. in κ, π, v decr. in p, v_0 incr. in κ, π, v decr. in p, v_0, σ
- $\psi_{\alpha} = \frac{\pi s(1 - F_{\alpha}(\bar{\alpha}_s))}{s(\pi + F_{\alpha}(\bar{\alpha}_s)) + (1+\pi)(1-s)F_{\alpha}(\bar{\alpha}_{\bar{s}})}$
 $\Rightarrow$ incr. in s, p, v_0, σ decr. in κ, v and $\bar{\alpha}$ ambiguous in π

Consumer

- $\bar{\sigma} \equiv \frac{v_0 + p}{\psi_{\alpha} v} - \frac{\pi(1 - \psi_{\alpha})}{(1+\pi)\psi_{\alpha}}$
 $\Rightarrow$ incr. in κ decr. in ψ_{α}, s, σ

Firm

- $\bar{\kappa}_H \equiv \rho[\psi_{\eta}(1 - s)F_{\alpha}(\bar{\alpha}_{\bar{s}}) - (1 - \psi_{\eta})s](p - c)$
 $\Rightarrow$ incr. in ρ decr. in c

ML-Equilibrium vs HML-Equilibrium

Influencer

- Uninformed influencer less likely to endorse: $\bar{\alpha}_s^{ML} < \bar{\alpha}_s^{HML}$
- Informed influencer unchanged: $\bar{\alpha}_s^{ML} = \bar{\alpha}_s^{HML}$

Consumer

- More parasociality needed to follow endorsements: $\bar{\sigma}^{ML} > \bar{\sigma}^{HML}$
- Value of genuine recommendation increases

Firm

- Changes in demand and threshold only through $\bar{\alpha}, \bar{\sigma}$
- Anticipating only *ML*-endorsements reduces profitability: $\bar{\kappa}_H^{ML} < \bar{\kappa}_H^{HML}$
 $\Rightarrow$ since $\bar{\kappa}_H^{ML} < \bar{\kappa}_H^{HML} \rightarrow$ multiplicity of HML/ML equilibrium

Consumer Surplus

- Same limitations for welfare as before

- Compare net consumer surplus

$$\begin{array}{ll}
 r_g: & [\sigma + (1 - \sigma)\pi]v - p - v_0 \quad \text{with prob. } (1 - \rho)s\pi \\
 r_p|\bar{s}: & \frac{\pi}{1+\pi}v - p - v_0 \quad \text{with prob. } \rho(1 - s)\psi_\eta F_\alpha(\bar{\alpha}_{\bar{s}}) \\
 r_p|0: & (1 - \sigma)\pi v - p - v_0 \quad \text{with prob. } \rho s(1 - \pi)\psi_\eta F_\alpha(\bar{\alpha}_s) \\
 r_p|v: & \sigma v - p - v_0 \quad \text{with prob. } \rho s\pi\psi_\eta
 \end{array}$$

⇒ Again, CS lower with endorsements & effect of ψ_η ambiguous

⇒ Under multiplicity: $CS^{ML} < CS^{HML}$ (payoffs, ψ_η , $\bar{\alpha}$ decreasing)

Example: ML-Equilibrium

Parameters

- $v = 8, v_0 = 1, p = 4, \kappa = 0.2, \pi = 0.6$
- $\rho = 0.9, s = 0.3, \sigma = 0.7$
- $\alpha \sim U[0, 1] \quad \sigma_\eta = \eta + (1 - \eta)\sigma, \eta \sim U[0, 1]$

Equilibrium Quantities

- $\bar{\alpha}_{\bar{s}} \approx 0.05 \quad \bar{\alpha}_s \approx 0.03$
 $\Rightarrow \psi_\alpha \approx 0.72$
- $\bar{\sigma} \approx 0.72$
 $\Rightarrow \psi_\eta \approx 0.93$
- $\bar{\kappa}_H \approx 0.04 \quad \bar{\kappa}_M \approx 0.11$

How to best visualize Examples?

The other ML-Endorsement Equilibrium

- Two (mutually exclusive) equilibria classes dependent on $\sigma v - p \gtrless v_0$
 $\Rightarrow$ focus now on $\sigma v - p > v_0$ [some $v_i = v$ get not recommended]

ML-Endorsement Equilibrium (Non-Aligned Influencers)

For intermediate κ there exists an equilibrium with endorsements in which

- (a) Only firm types $q = M, L$ decide to use endorsements at cost κ .
- (b) There exist thresholds $\bar{\alpha}_0$, $\bar{\alpha}_v$ and $\bar{\alpha}_{\bar{s}}$ such that an influencer
 - i. with $\alpha \leq \bar{\alpha}_{\bar{s}}$ endorses the product unseen.
 - ii. with $\alpha \leq \bar{\alpha}_0$ endorses any product and $\alpha < \bar{\alpha}_v$ only $v_i = v$ products.
- (c) All consumers follow genuine recommendations.
- (d) Sufficiently parasocial consumers ($\sigma_\eta > \bar{\sigma}$) also follow endorsements.

Influencer Decision

- Similar to before, but with major difference at $v_\ell = v$
 $\Rightarrow$ additional info of endorsement offer now harmful
- Three influencer types, accept endorsement iff
 - $\bar{s} : \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \frac{\pi}{1+\pi}v + p + v_0}$ [unchanged] $s|0 : \bar{\alpha}_0 \equiv \frac{\kappa}{\kappa - (1-\sigma)\pi v + p + v_0}$ [unchanged]
 - $s|v : (1 - \alpha)\kappa + \alpha[\sigma v - p - v_0] \geq 0$
 $\Rightarrow$ since $\sigma v - p < v_0$, now $\alpha < \bar{\alpha}_v \equiv \frac{\kappa}{\kappa - \sigma v + p + v_0}$
- $\Rightarrow$ One can show $\bar{\alpha}_v > \bar{\alpha}_0$ and $\bar{\alpha}_{\bar{s}} > \bar{\alpha}_0$
- Modified aligned paid recommendation probability $\psi_\alpha = \frac{s(F_\alpha(\bar{\alpha}_v) - F_\alpha(\bar{\alpha}_0))}{sF_\alpha(\bar{\alpha}_v) + (1-s)F_\alpha(\bar{\alpha}_{\bar{s}})}$
- $\Rightarrow$ Otherwise consumer problem unchanged: $\bar{\sigma} \equiv \frac{v_0 + p}{\psi_\alpha v} - \frac{\pi(1 - \psi_\alpha)}{(1 + \pi)\psi_\alpha}$

Firm Decision

- Firm endorses if costs are sufficiently small $\kappa \leq (D_E^q - D_0^q)(p - c)$
 $\Rightarrow$ effective demand with (D_E^q) and without (D_0^q) endorsements depends on v_ι
- Adjust endorsement demand D_E^q for abstaining influencer ($\alpha > \bar{\alpha}_v$)

v_ι	D_0	D_E
v	s	$\rho(1 - s)\psi_\eta F_\alpha(\bar{\alpha}_s) - \rho s(1 - \psi_\eta)F_\alpha(\bar{\alpha}_v) - \rho s(1 - F_\alpha(\bar{\alpha}_v)) + s$
0	0	$\rho(1 - s)\psi_\eta F_\alpha(\bar{\alpha}_s) + \rho s\psi_\eta F_\alpha(\bar{\alpha}_s)$

- Note that absolute gains $(D_E^q - D_0^q)$ decrease with q_ι (and hence with q)
 $\Rightarrow$ adjust firm-type thresholds $\bar{\kappa}_H$ and $\bar{\kappa}_M$ accordingly

Non-Existence of L-Endorsement Equilibria

Non-Existence of L-Endorsement Equilibria

In the baseline, there cannot exist an equilibrium in which only the L -type uses endorsements while the other types rely solely on genuine recommendations.

- Equilibrium possible from firm side with sufficiently high κ
- Paid recommendations accepted by sufficiently selfish influencer (low $\bar{\alpha}$)
- paid recommendation credibly signals $q = L \Rightarrow v_\theta = 0$
 - $\Rightarrow$ no paid recommendation followed by consumers
 - $\Rightarrow$ no reason to endorse for L -type firm

A Sketch of the mL Mixed Endorsement Equilibrium

Mixed Endorsement Equilibria

In any equilibrium, at most one firm finds it optimal to mix at the same time.

- H -firm never endorses, L -firm always, M -firm with prob. ε
 $\Rightarrow$ expected value from endorsement: $\frac{\varepsilon\pi}{2\varepsilon\pi+1-\pi}v$
- Influencer decision: $\bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \frac{\varepsilon\pi}{2\varepsilon\pi+1-\pi}v+p+v_0}$, $\bar{\alpha}_0 \equiv \frac{\kappa}{\kappa - (1-\sigma)\frac{\varepsilon\pi}{\varepsilon\pi+1-\pi}v+p+v_0}$
 $\Rightarrow$ aligned endorsement prob.: $\psi_{\alpha} = \frac{\varepsilon\pi s(1-F_{\alpha}(\bar{\alpha}_{\bar{s}}))}{\varepsilon\pi s + (\varepsilon\pi+1-\pi)sF_{\alpha}(\bar{\alpha}_{\bar{s}}) + (2\varepsilon\pi+1-\pi)(1-s)F_{\alpha}(\bar{\alpha}_{\bar{s}})}$
- Consumer decision: $\psi_{\alpha}\sigma + (1-\psi_{\alpha})\frac{\varepsilon\pi}{2\varepsilon\pi+1-\pi} \geq \frac{v_0-p}{v} \Rightarrow \bar{\sigma} \equiv \frac{v_0+p}{\psi_{\alpha}v} - \frac{\varepsilon\pi(1-\psi_{\alpha})}{(2\varepsilon\pi+1-\pi)\psi_{\alpha}}$
- Firm decision: choose ε such that $\kappa = \bar{\kappa}_M(\varepsilon)$

What's ahead

- Influencer indirect utility conditional on following recommendation (status quo)?
- Continuous vs. binary parasocial state space?
- Characterize the prob. 0 mixed endorsement equilibria?
- ML-Endorsement Equilibrium
- Comparative Statics and Welfare/CS Analysis

- Introduce product categories and influencer “specialization” $\zeta \in K$
 $\Rightarrow$ affects σ and possibly ω_ζ $\rightarrow$ allows parasociality as spillover effects
- Endogenize ω (firm choice) $\rightarrow$ limited signalling
- Costly quality investigation for unknowing influencer

That's All Folks!

If you have questions, comments or feedback,
don't hesitate to contact me or stop by.

✉ julian.klix@uni-mannheim.de

🏢 L7 3-5, Room 3.43



Appendix: Influencer Decision with unconditional utility

- Recall influencer's utility: $u_{\mathcal{I}} = (1 - \alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - v_0]$
 - Three influencer types, accept endorsement iff
 - $\bar{s} : (1 - \alpha)\kappa + \alpha\psi_{\eta}[\pi v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \psi_{\eta}(\pi v - p - v_0)}$
 - $s|v : (1 - \alpha)\kappa + \alpha\psi_{\eta}[\sigma v + (1 - \sigma)\pi v - p - v_0] \geq \alpha[\sigma v + (1 - \sigma)\pi v - p - v_0]$
 $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}}^v \equiv \frac{\kappa}{\kappa + (1 - \psi_{\eta})(\sigma v + (1 - \sigma)\pi v - p - v_0)}$
 - $s|0 : (1 - \alpha)\kappa + \alpha\psi_{\eta}[(1 - \sigma)\pi v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_s^0 \equiv \frac{\kappa}{\kappa - \psi_{\eta}((1 - \sigma)\pi v - p - v_0)}$
- $\Rightarrow$ one can show: $\bar{\alpha}_{\bar{s}}^v > \bar{\alpha}_{\bar{s}} > \bar{\alpha}_s^0$